

Neuropsychiatry, functional disorders and the law

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What is neuropsychiatry?

- Overlap of disciplines of neurology and psychiatry
- Psychiatrists with interest (+/- training) in neurology
- Clinically:
 1. Psychiatric aspects of neurological disease: Parkinson's, Huntington's, MS, Tourette's
 2. Neurological presentations of psychiatric disorder: functional (conversion) disorder / other dissociative or somatoform conditions
- Teams work alongside other neuroscience disciplines such as neurology and neurosurgery, providing comprehensive care to patients

Neuropsychiatric health needs

- Three main sources of epidemiological data
 - neuropsychiatric conditions in neurosciences settings (prevalence 40-60%)
 - neuropsychiatric conditions in various individual neurological conditions (prevalence 30-50%)
 - organic presentations to general adult services (approx. 10%) requiring neuropsychiatric assessment and treatment

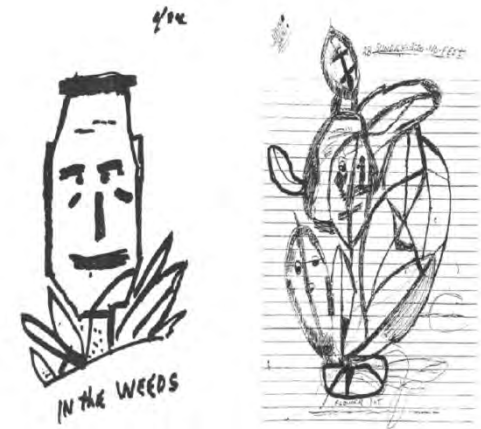


Figure 5-2. Illustrations by a man with Parkinson's disease of hallucinations induced by levodopa and selegiline.

Neuropsychiatric service demand

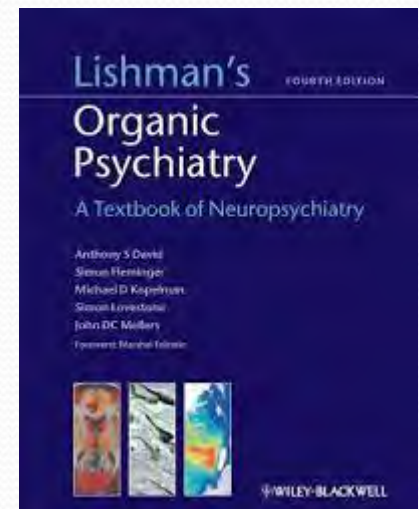
- National audit;- current service provision is patchy and limited to a few regional or national specialised centres
- Many existing services do not have adequate access to MDT staff or resources
- Inpatient services are available only at a few centres in England
- *Current wait in excess of 1 year for inpatient neuropsychiatry admission in London national centres*
- **In summary, there is a large unmet need for specialised neuropsychiatry**

Impact of the gap in provision

- In the absence of Tier 1 and 2 neuropsychiatry;
 - patients inappropriately placed in general psychiatric wards without expertise or multidisciplinary resources to manage them.
 - Alternatively they go to private/independent services, at significant cost to the NHS.
- Patients with FND attract excessive costs associated with repeated A&E presentations and ambulance costs, multiple investigations and extended bed occupancy.
- Quality of life strongly determined by co-morbid depression and anxiety in patients with epilepsy, stroke and Parkinson's disease (PD).

SLaM neuropsychiatry - history

- Early 1960s - Neuropsychiatry sub-specialty implemented by Sir Aubrey Lewis
- Alwyn Lishman became first UK Professor of Neuropsychiatry
- Epilepsy surgery pioneered at King's/Maudsley: “epilepsy ward” developed into Lishman Unit
- Specialist clinics developed; skilled nurse therapists trained
- Neuropsychiatry Liaison Service set up to serve the Regional Neurosciences Centre at Kings in the late 1980s’ early 90s.
- Neuropsychiatry established at STH.



Present

- SLaM Neuropsychiatry has changed from a loose collection of national and specialist services to be part of the Psychological Medicine CAG
- Also part of joint CAG/Institute of Neurosciences in Clinical Neurosciences KHP.
- Current service has an excellent national reputation; is a model for other regional units.
- Currently 8 consultants in service



Functional disorders; changing terminology

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- Hysteria
 - Conversion disorder. 'conversion' of unconscious psychological conflicts into physical symptoms
 - Dissociative disorder
 - Functional neurological disorder

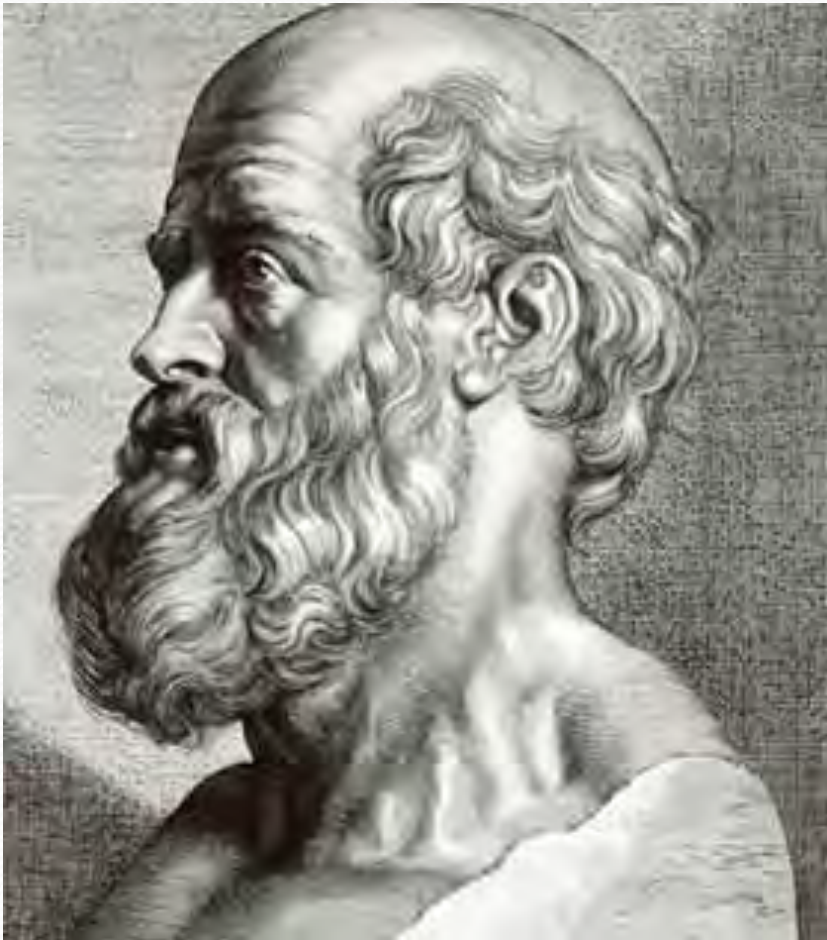
All imply:-

- psychological aetiology
(except hysteria!)

Differences:-

- psychological model / theory
- implications for *mechanism*

Hippocrates (5th Century BC) - Hysteria



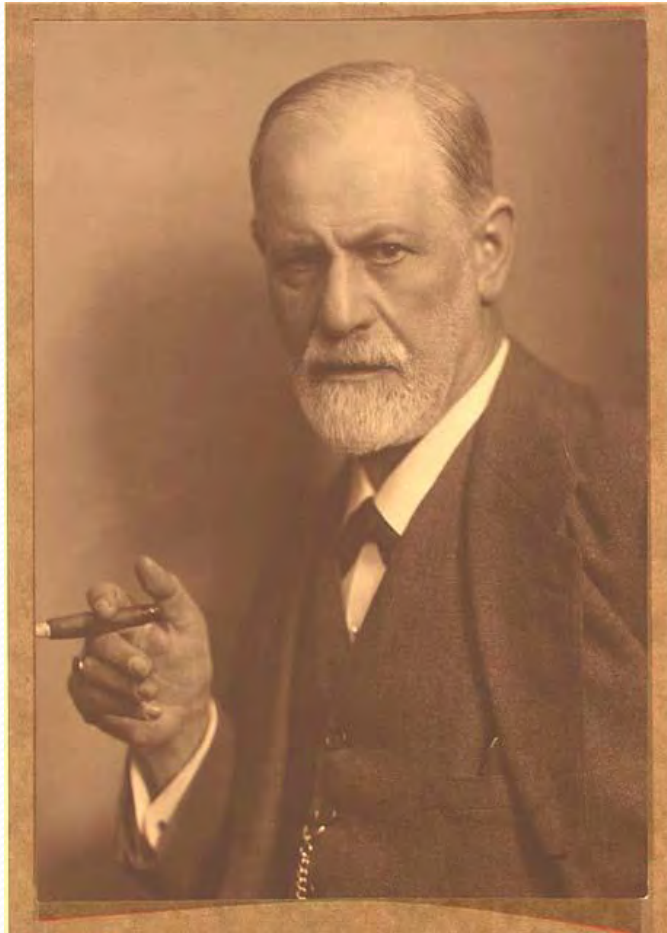
- Corpus Hippocraticum refers to various illness symptoms caused by movement of the uterus around the body
- Uterus becomes light and dry due to a lack of bodily fluids.
- “*Wandering womb*” theory persisted in European medicine for centuries
- In extreme cases a patient might have undergone surgical hysterectomy

Jean-Martin Charcot 1825-1893

- French neurologist at the Salpêtrière hospital
- reintroduced hysteria as a topic for scientific study
- believed hysteria was a neurological disorder
- 'hypnotisability' a clinical feature of hysteria
- influenced Freud



Sigmund Freud 1856-1939



- *Conversion disorder*
 - Unconscious 'conversion' of unresolvable psychological conflicts into physical symptoms.
 - Production of a physical symptom resolved the distress caused by repressed memories
 - Symbolism considered important

ICD-10 F44 Dissociative [conversion] disorders

- Mental states
 - F44.0 Dissociative amnesia
 - F44.1 Dissociative fugue
 - F44.2 Dissociative stupor
 - F44.3 Trance and possession disorders
- Physical symptoms
 - F44.4 Dissociative motor disorders
 - F44.5 Dissociative convulsions
 - F44.6 Dissociative anaesthesia and sensory loss
- F44.7 Mixed dissociative [conversion] disorders

What is dissociation?

- DEFINITION: loss of the normal integration between memories of the past, awareness of identity, immediate sensations, and *control of bodily movements*.
- *usually* we have conscious control over the memories and sensations that can be selected for immediate attention, and the movements that are to be carried out.
- in dissociative disorders the ability to exercise conscious and selective control is impaired

Everyday examples:

- Daydreaming
- Lapses when driving

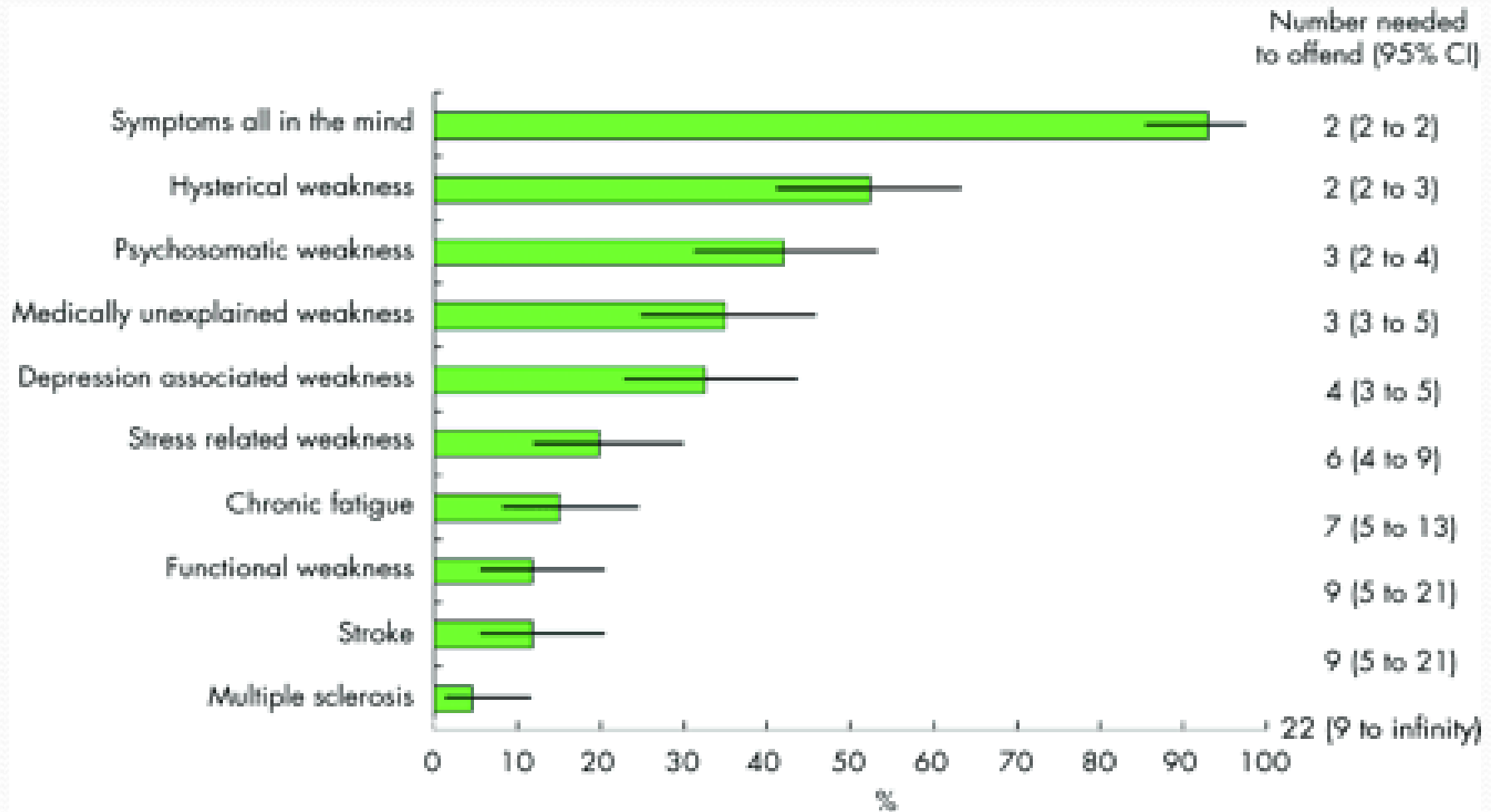
Overlap with:

- Normal experience
- Response to acute stress

Functional Neurological Disorder

- Nomenclature
- Currently fashionable term is *functional neurological disorder or FND*
- Common presentations are hyperkinetic disorders (tremors), weakness, dystonia, seizures
- Patients' support group – FNDhope
- Website;- www.neurosymptoms.org

The “Number needed to offend”



Modern concepts of aetiology

- Patient's ***consciousness*** of an affected symptom or capacity to move has been disturbed
- Disorder of ***attentional processes***– symptoms require attention and *vary with attention*
- Evidence of preserved function should not be ***misinterpreted*** as evidence of deliberate deception

Bayesian Inference Machine? – A Bayesian account of hysteria (Edwards MJ, Brain 2012)

- Balance between prior beliefs and sensory evidence mediated by attention, symptom expectations, physical and emotional experiences and beliefs about illness.
- Bayesian system because its hypotheses regarding the causes of sensory input at any hierarchical level are derived from both prior beliefs about the world and current sensory evidence
- *Primary failure* is the (autonomous) emergence of a percept or belief held with undue certainty (precision)
- *Secondary failure* is misattribution of agency

Neuropsychiatric assessment and diagnosis

- Full psychiatric/medical history – life events/trauma
- Physical examination
- Inconsistencies e.g. sudden onset of dystonia, entrainment of tremor, distraction, Hoover's sign
- Should be a positive diagnosis not a diagnosis of exclusion
- However, review by a specialist movement disorder neurologist / other relevant specialist is recommended

Interaction between physical and psychological disorder:

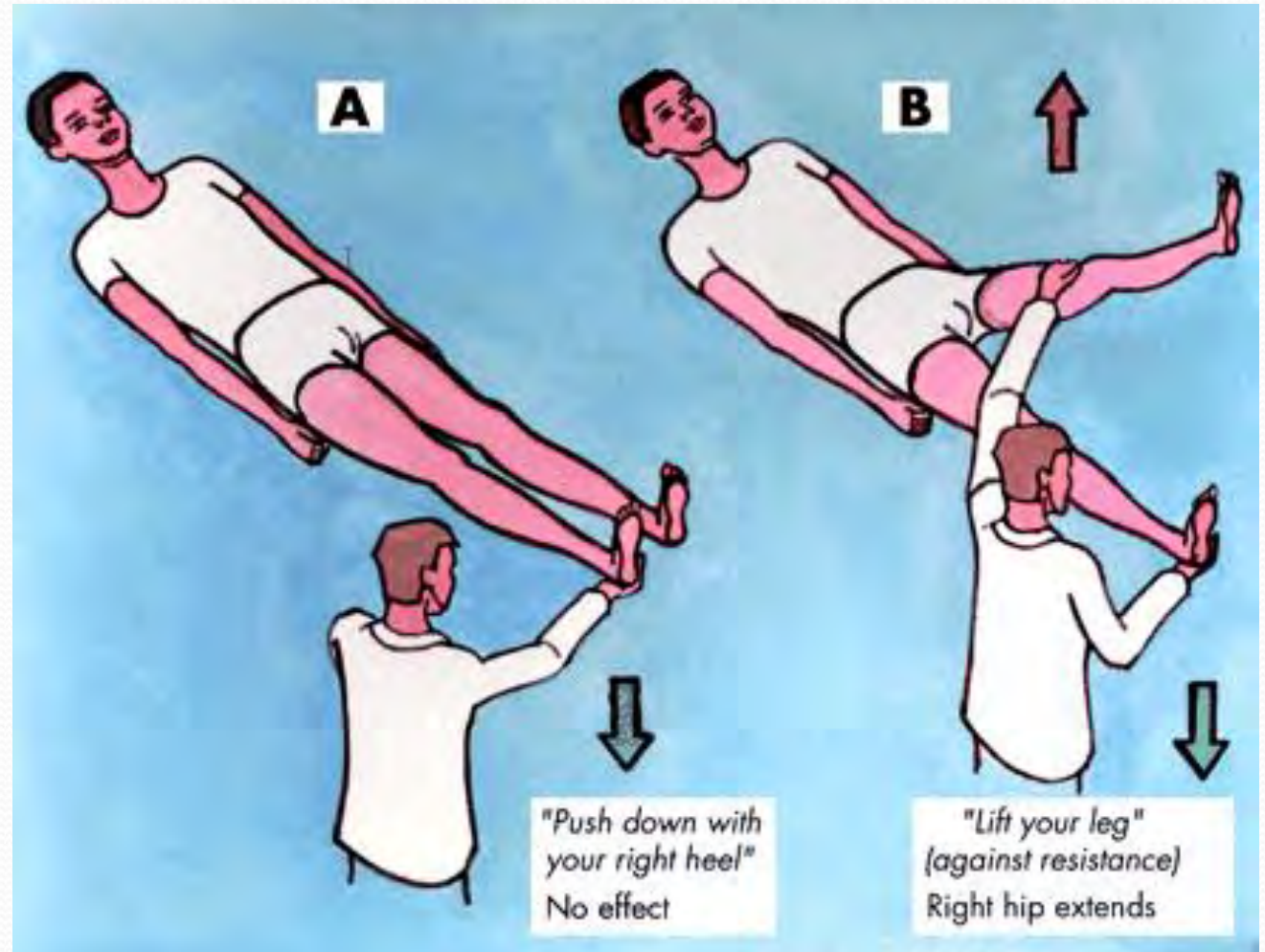
- Physical injury often precedes onset of conversion disorder (Stone, J 2009 J Psychosomatic Res)
- Physical disease often precedes onset of conversion disorder: 'functional overlay'
 - Comorbid epilepsy and conversion seizures common
 - Comorbid stroke / MS and conversion can occur
 - Neurobiological damage in mTBI and post-concussion syndrome

Incongruity with physical disease

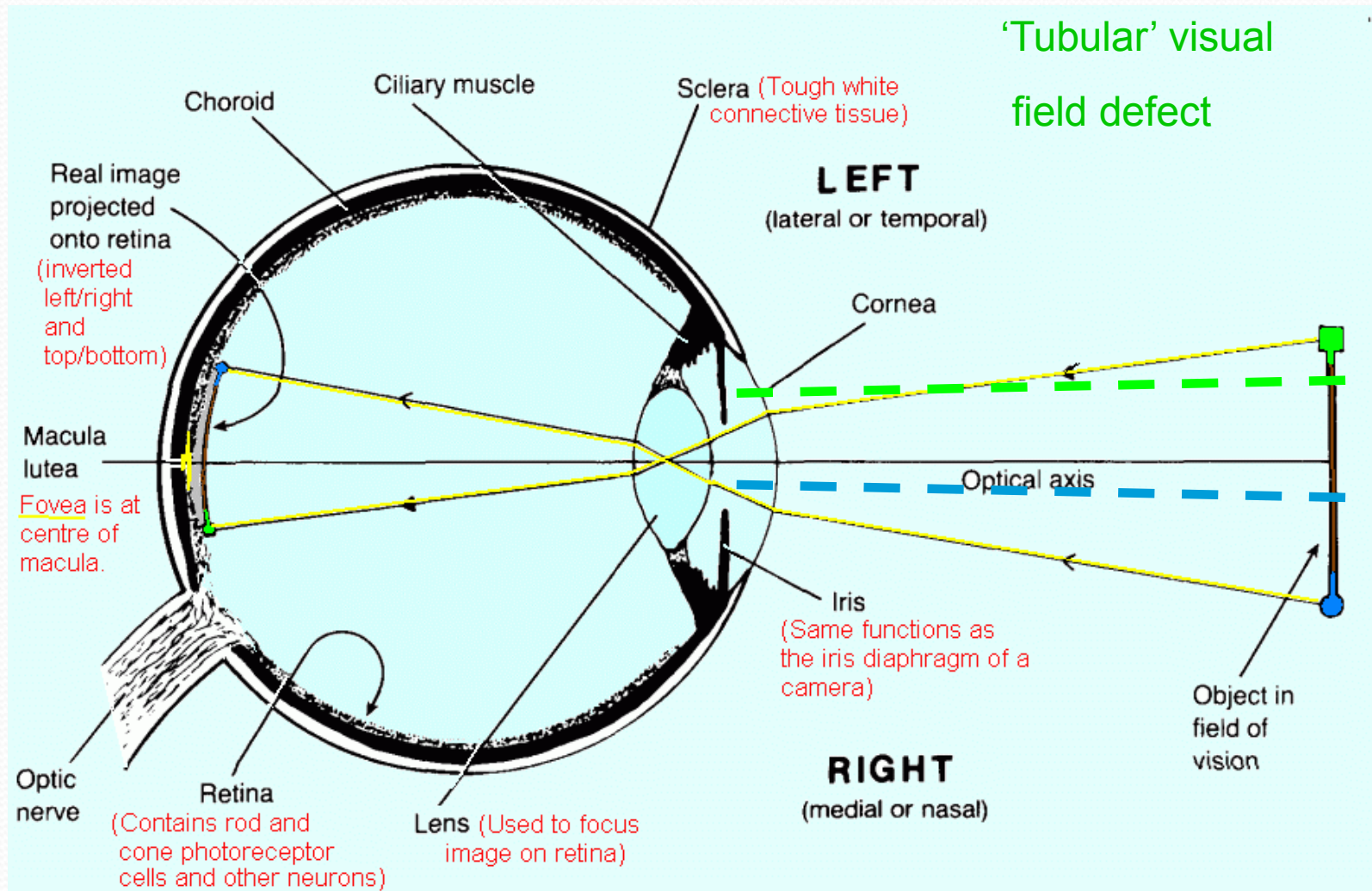
- Greater than expected inconsistency of symptoms / signs
- Not only *doesn't* fit with organic neurological disease patterns but 'at variance with anatomical / physiological patterns', i.e. *can't* fit neurological disease
- Better subconscious performance (compared to conscious performance)-observed strength / balance / vision better than on examination

Subconscious > conscious performance

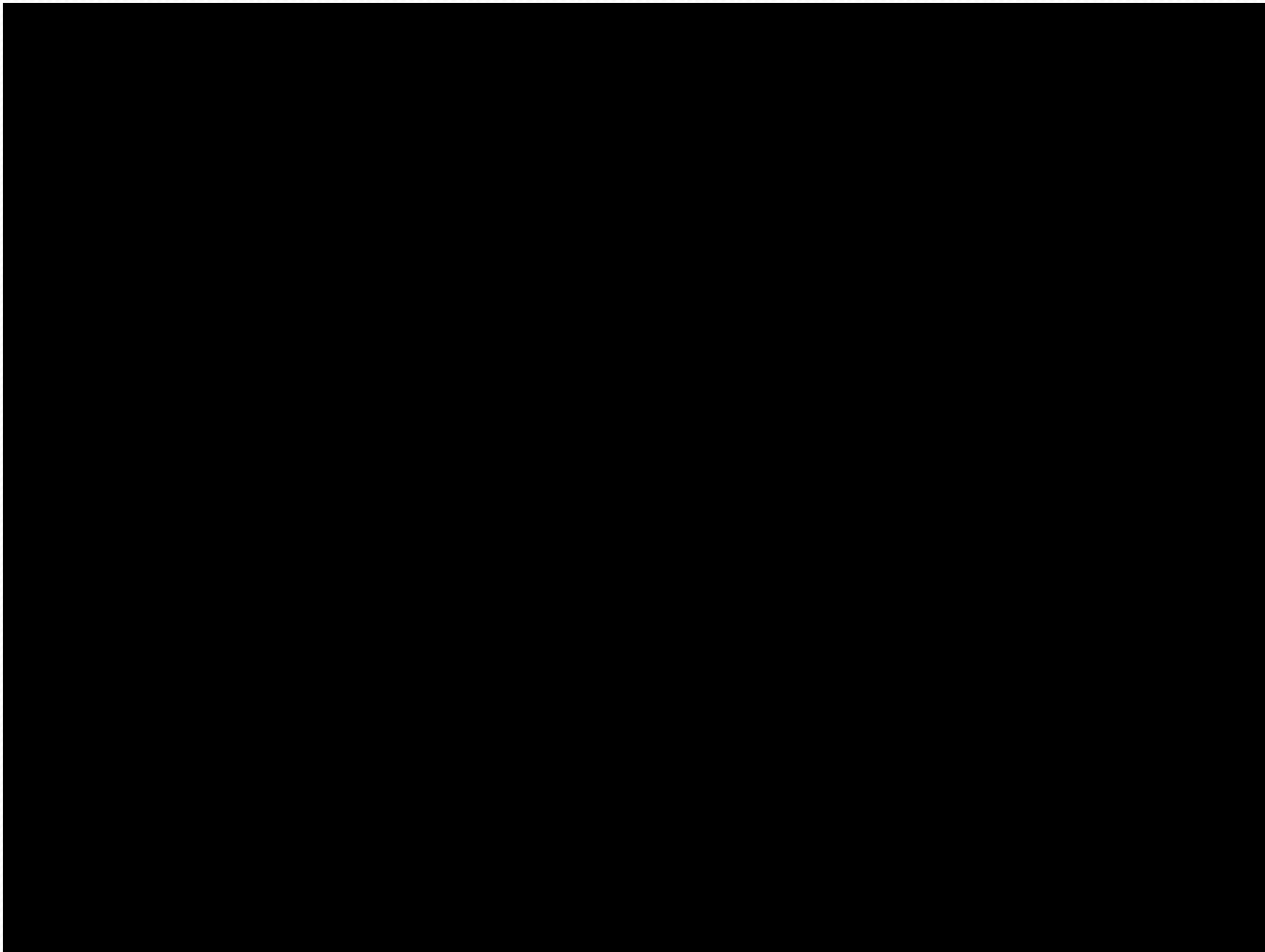
Hoover's sign:



Functional blindness; incongruity with physical disease



Entrainment of tremor



Attention effect!



Gait disorder



Post-injury



Functional neurological disorders-health needs

- One-third of neurology outpatients have symptoms rated as 'somewhat' or 'not at all' explained by organic disease.
- 20% of patients in apparent 'status epilepticus' and 1/7 'first fit' clinic pts have non-epileptic attacks
- Functional weakness; at least as common as MS, commonest non-stroke diagnosis in patients wrongly thrombolysed for presumed stroke.
- Of 3,781 neurology patients 5% had primary diagnosis of functional symptoms, such as non-epileptic attacks, functional weakness and movement disorders'

Treatment

- Combination of
 - CBT
 - Physiotherapy
 - Occupational therapy
 - Therapeutic environment
- Evidence
- Specialist inpatient treatment for severe motor conversion disorder: a retrospective comparative study. McCormack R et al. JNNP 2013

How to work with functional disorder patients

- Develop & maintain empathy for patients
 - ‘Walk around in their shoes’: would we be ‘normal’ if we had lived their lives?
 - Role of abuse / trauma
 - Learn how to manage personality disorder
- Develop interest in the aetiology
 - See patients as brain dysfunction
 - Issues of volition / free will / consciousness / hypnosis

mTBI / Post-concussion syndrome

- Non-specific symptoms
 - Headaches
 - Fatigue
 - Vertigo/dizziness
 - Irritability/emotional lability
 - Cognitive problems
 - Sleep disturbance
 - Anxiety and depression

Non-specific nature of symptoms

- Symptoms also found in other clinical and non-clinical groups
 - Healthy adults
 - Chronic pain patients
 - Spinal injury patients
 - Non brain injured trauma/orthopaedic patients
- Lack of specificity increases potential for misdiagnosis
- Prevalence estimates after mTBI 10-20%

Mild TBI / post-concussion syndrome

- Neuropsychological model of functional disability after mTBI (Kay et al 1992)
 - Neurobiological factors
 - Physical factors – pain etc.
 - Psychological distress–
 - create further cognitive compromise-cause attention/memory problems
 - leads to anxiety/avoidance/depression
 - Personality variables can affect the way an individual interprets symptoms/recovery
- *Psychological overlay intensifies and may become worse than initial injury*

Disability after brain injury: Personal and situational factors (Montgomery 1995)

Personal factors;

- Negative cognitions
- Physical symptoms
- fatigue

Situational factors;

- Rapid processing demands
- External distractions
- Need for complex attention

- Rehabilitation focused on working with patients to modify these factors
- Identification of non-organic components

Evidence for neurobiological causation for psychological syndromes in mTBI

- Depression- fronto-limbic subcortical structures involved in biological models of depression. Reduced activation in depressed concussed patients (Chen et al.)
- Atrophy in these regions correlates with depressive symptoms (Hudak et al)
- Reduced hippocampal volume in PTSD (Bremner JD)
- Brain injured rodents show increased fear responses (Reger et al.)
- Taken together these findings suggest that neurological abnormalities may predispose to emotional distress post TBI

Aetiology

- Lishman (1988): distinguished between ‘physiogenesis’ and ‘psychogenesis’
 - Acute symptoms=neurobiological; resolve with time
 - Persisting symptoms=psychological
- Silverberg (2011): more global role for psychological factors
 - Contribute on early stages
 - Persist in prolonged PCS
 - Predict late outcome of PCS

Predictors of PCS

- Pre-injury anxiety and acute PTSD (Meares et al)
- Pre-morbid psychiatric history, life stressors and pain (Posford et al)
- Coping style (emotion focused vs. problem focused) (Woodrome et al)
- Anxiety sensitivity (Bodily sensations related to anxiety are misattributed as a harmful causing more intense anxiety/fear) correlates with PCS symptoms (Wood et al)

Cognitive misattribution and expectancy effects

- Some pts anticipate PCS symptoms will be a natural part of their recovery
- ‘expectation as aetiology’ (Mittenberg)
- May attribute *any perceived difficulty* (i.e. headaches) to the injury
- Minimise other causes e.g. stress

'Good old days' bias

- 90 PCS; 177 controls (Iverson et al)
- PCS group asked to estimate pre-injury symptoms and endorse current symptoms; Control group asked to indicate current symptoms
- Concussed group reported fewer symptoms pre-injury than base rate in the healthy control group
- Education on base rates could be an important intervention during recovery

Treatment

- Graded return to activity
- Anxiety reduction techniques
- Cognitive-behavioural therapy of cognitive biases and misattribution
- Interventions for psychosocial factors complicating recovery
- Pharmacological intervention for refractory post-concussion depression (SSRIs)

Factitious disorder/malingering

- Factitious- **conscious** feigning for psychological gain (carer support, medical attention – Munchausen's syndrome)
- Malingering - **conscious** feigning for material/financial gain
- Inconsistency of symptoms does not equate to malingering (symptoms vary with attention, also self-deception occurs)
- Malingering is **not a medical diagnosis**

Further reading

- Neuropsychiatry workbook & DVD (Maudsley publications) available at Institute of Psychiatry library
- Conversion disorder: Review articles by Jon Stone (JNNP 2005 and Practical Neurology 2009) and Tim Nicholson (Psychiatry 2009)
- Info for patients at www.neurosymptoms.org
- Non-epileptic seizures: review articles by John Mellers or Markus Reuber
- Psychogenic memory disorders: review articles by Michael Kopelman or Adam Zeman